Supplementary material

Adjustment and external replication of drug–hearing-loss signals: a disproportionality analysis of FAERS and Canada Vigilance

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Table S1. READUS-PV checklist

Item	READUS-PV recommendation (abridged)	Where addressed
1a	Identify the study as a disproportionality analysis; name data type and database(s)	Title
1b	Name the adverse event(s) and/or drug(s)	Title (hearing loss); abstract
2a	Drugs, event, frequency, existing knowledge	Introduction, paragraph 1
2b	Rationale (routine PV, safety profile or pre-specified hypothesis)	Introduction, paragraphs 2–3
2c	Why ICSR databases and disproportionality suit the gap	Introduction, paragraphs 1–3
3	Objectives: event(s), drug(s), reference group, pre-specified hypotheses	Introduction, last paragraph; Methods 2.4
4a	Identify design and data type	Methods 2.1
4b	Outline primary and sensitivity analyses, case-by-case analysis	Methods 2.4–2.8; PROTOCOL.md
5a	Database names, custodians, coverage, thesaurus	Methods 2.1 (FDA, Health Canada; MedDRA PTs)
5b	Extraction dates; pre-processing and exclusions	Methods 2.1–2.2; Results 3.1; Figure S1
6a	Study population and restrictions	Methods 2.3–2.4
6b	Key variables	Methods 2.3
6c	Grouping of drugs/events; ingredients vs trade names; drug role	Methods 2.2–2.3
6d	Additional data sources and how they interact with ICSRs	Methods 2.5 (US labels), 2.8 (PubMed)
7a	Descriptive analyses	Table S4
7b	Disproportionality measures and thresholds	Methods 2.4
7c	Sensitivity analyses and confounding control	Methods 2.4, 2.6, 2.7; Results 3.3–3.4
7d	Variables and criteria for case-by-case analysis	Methods 2.8; Table S3
7e	Statistical methods for other data sources	Methods 2.5 (AUC, bootstrap), 2.6
8a	Number of ICSRs at each stage with reasons	Results 3.1; Figure S1
8b	Demographic and clinical characteristics of cases	Table S4
9	All results with confidence intervals; sensitivity analyses	Tables 1–4, S2, S5, S6; Figures 1–2
10	Case-by-case analysis	Results 3.6; Table 4, S3
11	Key results in context; expected vs emerging signals	Discussion, paragraphs 1–2
12a	External validity	Discussion; Limitations
12b	Relevance to clinical practice	Discussion, paragraph 4
12c	Further study designs	Discussion, paragraph 4; Conclusions
13	Limitations; disproportionality cannot prove causation or measure incidence	Limitations
14a	Funding	Declarations
14b	Conflicts of interest	Declarations
14c	Institutional approval	Declarations (not applicable)
14d	Data, code (software versions) and protocol availability	Methods 2.9; Declarations

Table S2. Established ototoxic drugs: crude and adjusted estimates (hearing loss)

FAERS crude: all reports; FAERS adjusted: Mantel–Haenszel OR in the restricted population (Methods); Canada: crude ROR, all reports.

Ingredient	FAERS cases	FAERS ROR (95% CI)	Adj. cases	FAERS adj. OR (95% CI)	Canada cases	Canada ROR (95% CI)
Gentamicin	315	9.66 (8.63–10.81)	181	11.37 (9.79–13.20)	131	33.60 (27.98–40.34)
Tobramycin	367	5.36 (4.83–5.94)	248	4.82 (4.25–5.47)	31	17.10 (11.89–24.59)
Amikacin	833	16.83 (15.69–18.07)	472	12.70 (11.57–13.95)	35	145.93 (97.41–218.60)
Streptomycin	18	5.76 (3.61–9.20)	12	7.92 (4.47–14.05)	9	6.28 (3.24–12.17)
Cisplatin	1206	4.61 (4.35–4.88)	730	4.27 (3.96–4.59)	138	31.21 (26.13–37.29)
Carboplatin	761	1.96 (1.83–2.11)	251	0.82 (0.72–0.93)	112	19.09 (15.74–23.17)
Oxaliplatin	121	0.44 (0.37–0.53)	106	0.46 (0.38–0.56)	1	0.14 (0.02–1.02)
Vancomycin	365	2.25 (2.03–2.49)	215	2.06 (1.80–2.35)	60	9.25 (7.14–11.98)
Furosemide	412	1.73 (1.57–1.91)	255	1.31 (1.16–1.48)	61	10.18 (7.87–13.16)
Hydroxychloroquine	556	2.10 (1.93–2.28)	327	1.73 (1.55–1.93)	88	2.21 (1.79–2.73)
Quinine	68	5.83 (4.58–7.42)	41	6.10 (4.47–8.31)	4	6.10 (2.27–16.42)
Deferoxamine	51	8.97 (6.78–11.86)	50	11.78 (8.87–15.65)	34	36.29 (25.40–51.86)
Sildenafil	894	3.99 (3.73–4.26)	725	3.93 (3.65–4.24)	32	4.19 (2.95–5.94)
Tadalafil	720	4.52 (4.20–4.87)	594	4.22 (3.89–4.58)	35	4.19 (3.00–5.85)
Vardenafil	6	1.31 (0.59–2.93)	4	0.93 (0.35–2.49)	4	17.01 (6.21–46.61)
Teprotumumab	1133	59.90 (56.11–63.95)	859	59.76 (55.40–64.47)	1	–

Figure S1. Report flow

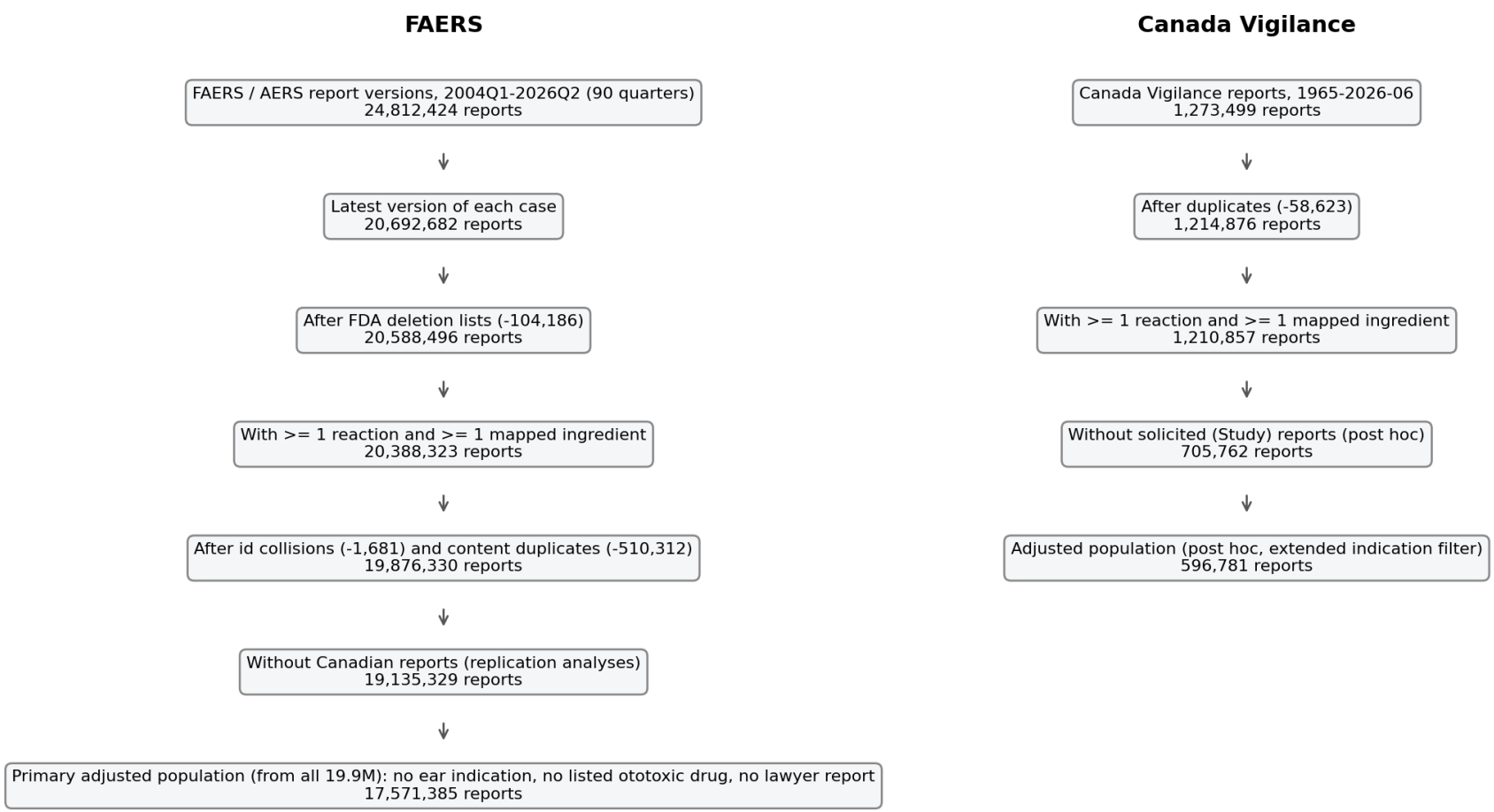


Table S3. Case-level review of robust signals absent from all US labels (post hoc)

Rubric, first applicable category: K established ototoxic class or association documented by a prospective/cohort series; A report cluster ($\geq$50% of cases in reports listing $\geq$15 drugs); B $\geq$50% of cases co-name an established ototoxic drug; C stimulated or solicited reporting; D main indication associated with the event; E none. Cat. 1: author; Cat. 2: second, AI-based assessor, blind to Cat. 1, before the active-comparator rule (agreement 19/22, $\kappa = 0.82$). Category D also requires the active-comparator (within-indication) ROR lower bound ≤ 1. Profiles: FAERS cases without Canadian reports.

Drug	Event	Cat. 1	Cat. 2	Cases	Top country	Top year	Consumer	Median drugs	≥15 drugs	Ototoxic co-drug	Within-indication ROR	Note
Rofecoxib	HL	C	C	243	US 64%	2006 42%	7%	6	14%	33%		litigation-era reporting (2004-2
Tozinameran	HL	C	K	57	US 40%	2022 28%	72%	6	26%	5%		mass vaccination; cohort evic conflicting (Israel SIR 1.35, P 35201275; Denmark null, P 37288514)
Elasomeran	HL	C	C	20	US 85%	2023 35%	75%	8	35%	20%		mass vaccination; media atte to hearing effects
Etanercept	HL	D	D	2095	US 80%	2016 20%	49%	1	3%	4%	0.93 (0.87-1.00)	rheumatoid arthritis assoc with SNHL (PMID 32488 used for autoimmune inner disease
Insulin Aspart	HL	D	D	250	US 85%	2025 17%	82%	2	5%	2%	1.17 (1.00-1.37)	diabetes associated with he impairment (meta-analysis, P 23150692)
Tofacitinib	HL	E	D	1728	US 94%	2022 19%	82%	1	4%	7%	2.24 (2.06-2.45)	persists against other RA d 82 % consumer reports; rhe toid arthritis associated with S (PMID 32488430)
Pregabalin	HL	E	E	1703	US 86%	2021 23%	62%	2	12%	11%	3.27 (3.04-3.52)	alpha-2-delta ligand; one report of reversible loss (P 32297964); diabetes/neurop and age (median 71) confo labels list tinnitus and hyperac not hearing loss
Amlodipine	HL	E	D	792	US 29%	2021 15%	70%	4	9%	14%	1.89 (1.69-2.10)	labels list tinnitus; VA cohort f lower SNHL risk with calc channel blockers (PMID 34855 hypertension associated with ing loss (meta-analysis, P 39839866)
Glycopyrrolate	HL	E	D	315	US 75%	2025 27%	50%	3	3%	4%	1.65 (1.27-2.14)	persists against other COPD d COPD associated with hearing (PMID 38828668)
Travoprost	HL	E	D	140	US 61%	2019 23%	75%	1	0%	3%	2.43 (1.80-3.29)	persists against other glauc drugs; median age 80; coma associated with SNHL (P 31935130)
Candesartan	Tin	A	A	422	GB 80%	2023 54%	38%	42	83%	34%		83% of cases in reports listing 15 drugs
Propranolol	Tin	A	A	189	PT 50%	2024 21%	24%	71	66%	66%		66% of cases in reports listing 15 drugs
Tozinameran	Tin	A	A	109	GB 52%	2023 38%	46%	28	51%	48%		51% of cases in reports listing 15 drugs; mass vaccination; hort evidence conflicting (Israe 1.35, PMID 35201275; Den null, PMID 37288514)
Indapamide	Tin	A	A	44	PL 50%	2018 23%	18%	15.5	55%	59%		55% of cases in reports listing 15 drugs
Finasteride	Tin	C	C	293	GB 38%	2024 17%	60%	1	0%	1%		post-finasteride syndrome advo reporting

Rofecoxib	Tin	C	C	189	US 58%	2006 47%	8%	6	21%	26%		litigation-era reporting (2004-2
Elasomeran	Tin	C	C	13	US 54%	2023 38%	69%	8	31%	23%		mass vaccination; media atte to hearing effects
Galcanezumab	Tin	D	E	96	US 94%	2020 28%	88%	1	0%	1%	0.82 (0.63-1.06)	migraine associated with t tus (systematic review, P 36459425; a meta-analysis f low evidence, PMID 36380120
Chlorthalidone	Tin	D	E	10	US 70%	2025 20%	90%	4	0%	0%	1.04 (0.43-2.51)	hypertension associated tinnitus (meta-analysis, P 26881064)
Doxycycline	Tin	E	E	220	US 43%	2024 13%	48%	3	6%	23%	3.63 (2.98-4.42)	pulsatile tinnitus can accom labelled drug-induced intracr hypertension (PMID 38516 but only 15/220 cases co-r intracranial hypertension or p loedema
Liothyronine	Tin	E	D	40	US 57%	2023 35%	78%	5.5	32%	2%	2.10 (1.31-3.36)	persists against other hyp roidism drugs; 32 % drug- reports (UK cluster overlap); pothyroidism associated with t tus (PMID 35233053)
Paromomycin	Tin	K	K	3	US 67%	2017 33%	67%	1	0%	0%		on the a-priori ototoxic list (ar glycoside); oral aminoglyc poorly absorbed; 3 cases

Table S4. Characteristics of FAERS hearing-loss and tinnitus reports and all other reports (READUS-PV item 8b)

The hearing-loss and tinnitus groups overlap: 6,722 reports contain both events and are counted in both columns. "All other reports" are reports with neither event, so the three columns sum to more than the 19,876,330 reports analysed.

	Hearing loss	Tinnitus	All other reports
Reports, n	83,883	42,900	19,756,269
Age, median (years)	67	55	58
Age			
<18 (%)	2.4	1.2	3.5
18-44 (%)	6.9	16.3	12.5
45-64 (%)	15.6	30.0	20.2
≥65 (%)	30.6	17.4	20.7
unknown (%)	44.5	35.1	43.1
Sex			
female (%)	52.9	58.3	52.7
male (%)	39.3	32.2	34.3
unknown (%)	7.8	9.5	13.0
Reporter			

consumer (%)	60.0	53.7	44.2
physician (%)	16.2	16.9	21.6
pharmacist (%)	3.2	4.7	6.1
other health professional (%)	12.6	14.2	18.6
lawyer (%)	0.8	1.8	2.5
unknown (%)	7.2	8.7	6.9
Country			
United States (%)	69.9	64.0	71.3
other (%)	28.6	32.8	27.0
unknown (%)	1.5	3.2	1.7
FDA receipt			
2004-2012 (%)	13.6	21.2	15.6
2013-2018 (%)	30.7	34.0	31.8
2019-2026 (%)	55.7	44.8	52.6

Table S5. All robust signals (post hoc: adjusted FAERS signal replicated adjusted-to-adjusted in Canada Vigilance without solicited reports)

FAERS: adjusted MH OR (extended indication filter, Canadian reports excluded); Canada: adjusted MH OR; Label: event on any US label of a product containing the drug.

Ingredient	Event	FAERS adj. cases	FAERS adj. OR (95% CI)	Canada adj. cases	Canada adj. OR (95% CI)	Label
Amikacin	HL	454	12.71 (11.55–13.98)	20	286.54 (160.73–510.83)	yes
Amlodipine	HL	675	2.39 (2.21–2.57)	16	2.70 (1.65–4.44)	no
Amphotericin B	HL	43	1.53 (1.14–2.07)	3	6.30 (2.01–19.75)	yes
Azithromycin	HL	404	5.15 (4.66–5.68)	30	11.05 (7.65–15.96)	yes
Bupropion	HL	219	1.15 (1.01–1.32)	21	2.46 (1.59–3.80)	yes
Ciprofloxacin	HL	400	2.94 (2.66–3.24)	35	4.47 (3.19–6.27)	yes
Cisplatin	HL	688	4.30 (3.98–4.64)	47	20.99 (15.49–28.45)	yes
Clarithromycin	HL	196	3.40 (2.95–3.91)	21	5.45 (3.52–8.43)	yes
Deferasirox	HL	290	4.02 (3.58–4.52)	6	4.29 (1.91–9.66)	yes
Deferoxamine	HL	35	9.60 (6.85–13.45)	24	81.44 (51.89–127.82)	yes
Elasomeran	HL	16	2.36 (1.44–3.86)	16	3.30 (2.01–5.44)	no
Erythromycin	HL	57	2.29 (1.76–2.97)	20	7.74 (4.90–12.22)	yes
Etanercept	HL	2003	1.15 (1.10–1.21)	49	2.28 (1.71–3.05)	no
Ethacrynic Acid	HL	4	13.94 (5.07–38.28)	5	109.98 (41.12–294.17)	yes
Furosemide	HL	209	1.19 (1.04–1.36)	14	4.42 (2.59–7.53)	yes
Gentamicin	HL	168	11.96 (10.24–13.97)	90	65.05 (50.65–83.55)	yes
Glycopyrrolate	HL	303	3.08 (2.74–3.45)	5	4.40 (1.82–10.67)	no
Hydroxychloroquine	HL	133	1.80 (1.51–2.13)	63	2.03 (1.57–2.62)	yes
Insulin Aspart	HL	243	2.27 (2.00–2.58)	5	2.69 (1.11–6.50)	no
Moxifloxacin	HL	132	2.12 (1.78–2.51)	23	6.87 (4.52–10.43)	yes
Neomycin	HL	52	4.97 (3.77–6.54)	6	8.70 (3.85–19.66)	yes
Peginterferon Alfa-2B	HL	64	1.54 (1.20–1.96)	7	4.57 (2.15–9.70)	yes
Piroxicam	HL	15	2.20 (1.32–3.65)	3	5.65 (1.80–17.75)	yes
Pregabalin	HL	1497	3.25 (3.09–3.43)	15	2.06 (1.24–3.44)	no

Ramipril	HL	82	1.27 (1.02–1.58)	10	2.34 (1.25–4.37)	yes
Ribavirin	HL	323	1.25 (1.12–1.40)	11	1.98 (1.09–3.61)	yes
Rofecoxib	HL	143	2.39 (2.02–2.82)	10	4.42 (2.35–8.31)	no
Sildenafil	HL	705	4.04 (3.74–4.35)	23	5.44 (3.57–8.28)	yes
Streptomycin	HL	12	9.18 (5.17–16.29)	6	8.80 (3.89–19.93)	yes
Tadalafil	HL	587	4.42 (4.07–4.79)	17	4.43 (2.71–7.22)	yes
Terbinafine	HL	99	3.29 (2.70–4.01)	5	3.09 (1.28–7.48)	yes
Tobramycin	HL	232	4.82 (4.23–5.49)	17	23.12 (14.12–37.86)	yes
Tofacitinib	HL	1608	3.27 (3.11–3.45)	53	1.39 (1.05–1.84)	no
Tozinameran	HL	46	2.36 (1.76–3.15)	109	7.04 (5.70–8.69)	no
Travoprost	HL	135	3.71 (3.13–4.40)	3	4.96 (1.58–15.54)	no
Vancomycin	HL	209	2.21 (1.93–2.54)	42	12.16 (8.87–16.67)	yes
Acetazolamide	Tin	12	2.96 (1.68–5.23)	3	5.11 (1.63–16.07)	yes
Amikacin	Tin	233	17.86 (15.66–20.37)	3	21.25 (6.56–68.80)	yes
Amitriptyline	Tin	72	1.68 (1.33–2.12)	11	1.96 (1.08–3.56)	yes
Amlodipine	Tin	445	3.06 (2.79–3.36)	16	2.50 (1.52–4.10)	yes
Aspirin	Tin	615	3.01 (2.78–3.26)	118	7.51 (6.19–9.10)	yes
Atenolol	Tin	83	2.91 (2.35–3.62)	8	3.91 (1.94–7.88)	yes
Azithromycin	Tin	257	5.76 (5.09–6.52)	22	6.00 (3.92–9.19)	yes
Bisoprolol	Tin	321	13.85 (12.39–15.49)	4	3.29 (1.23–8.83)	yes
Bupropion	Tin	1058	7.90 (7.42–8.41)	59	4.51 (3.47–5.87)	yes
Buspirone Hydrochloride	Tin	44	3.84 (2.85–5.16)	4	8.70 (3.20–23.61)	yes
Candesartan	Tin	277	11.72 (10.40–13.22)	15	2.80 (1.68–4.68)	no
Celecoxib	Tin	154	1.67 (1.42–1.95)	20	1.70 (1.09–2.65)	yes
Chlorthalidone	Tin	10	1.94 (1.04–3.62)	4	4.57 (1.70–12.29)	no
Choline	Tin	4	23.45 (8.56–64.21)	4	3.98 (1.48–10.70)	yes
Ciprofloxacin	Tin	969	12.72 (11.92–13.57)	52	4.98 (3.77–6.58)	yes
Cisplatin	Tin	233	2.33 (2.05–2.65)	17	5.81 (3.58–9.43)	yes
Citalopram	Tin	179	2.37 (2.05–2.75)	12	1.77 (1.00–3.13)	yes
Clonazepam	Tin	254	2.99 (2.64–3.38)	16	3.58 (2.18–5.88)	yes
Cyclobenzaprine	Tin	33	1.57 (1.12–2.22)	4	2.96 (1.10–7.95)	yes
Desvenlafaxine	Tin	226	4.74 (4.15–5.40)	9	4.16 (2.14–8.06)	yes
Dexchlorpheniramine	Tin	8	4.34 (2.16–8.71)	5	2.80 (1.16–6.78)	yes
Diltiazem	Tin	48	1.70 (1.28–2.26)	11	4.05 (2.23–7.38)	yes
Doxycycline	Tin	159	4.54 (3.88–5.31)	7	2.36 (1.12–4.97)	no
Duloxetine	Tin	1718	12.40 (11.80–13.04)	32	4.16 (2.92–5.92)	yes
Elasomeran	Tin	10	3.62 (1.94–6.74)	21	3.50 (2.26–5.43)	no
Felodipine	Tin	220	73.24 (63.47–84.52)	3	5.34 (1.70–16.75)	yes
Finasteride	Tin	286	9.73 (8.63–10.95)	5	3.04 (1.26–7.37)	no
Gabapentin	Tin	341	2.02 (1.82–2.25)	25	4.21 (2.83–6.28)	yes
Galcanezumab	Tin	94	1.86 (1.52–2.28)	5	16.70 (6.73–41.48)	no
Gentamicin	Tin	45	5.75 (4.28–7.72)	23	8.48 (5.55–12.97)	yes
Ibuprofen	Tin	289	1.51 (1.34–1.69)	36	2.89 (2.07–4.04)	yes
Indapamide	Tin	18	2.05 (1.29–3.26)	6	3.44 (1.53–7.71)	no
Indomethacin	Tin	15	2.54 (1.53–4.22)	9	2.68 (1.39–5.20)	yes
Irbesartan	Tin	73	3.15 (2.50–3.97)	6	2.26 (1.01–5.07)	yes
Isotretinoin	Tin	212	2.69 (2.35–3.08)	10	3.17 (1.69–5.94)	yes
Lidocaine	Tin	90	1.67 (1.35–2.05)	12	2.85 (1.61–5.04)	yes
Liothyronine	Tin	38	4.90 (3.56–6.75)	3	16.73 (5.22–53.59)	no
Mefloquine	Tin	85	27.80 (22.24–34.74)	8	14.52 (7.08–29.79)	yes
Minocycline	Tin	49	4.01 (3.02–5.31)	7	3.49 (1.65–7.36)	yes
Moxifloxacin	Tin	151	3.51 (2.99–4.13)	21	4.65 (3.01–7.18)	yes
Naproxen	Tin	445	3.05 (2.77–3.35)	23	2.42 (1.60–3.66)	yes

Nifedipine	Tin	24	1.51 (1.01–2.26)	7	2.95 (1.40–6.22)	yes
Norfloxacin	Tin	7	4.63 (2.19–9.75)	4	6.03 (2.23–16.29)	yes
Nortriptyline	Tin	29	2.80 (1.94–4.04)	6	3.96 (1.76–8.87)	yes
Paromomycin	Tin	3	27.75 (8.63–89.29)	3	262.76 (53.09–1300.46)	no
Paroxetine	Tin	478	5.19 (4.74–5.69)	38	5.12 (3.70–7.11)	yes
Penicillin G Procaine	Tin	5	7.93 (3.27–19.22)	5	10.12 (4.12–24.86)	yes
Perindopril Erbumine	Tin	6	2.33 (1.05–5.20)	12	3.59 (2.02–6.37)	yes
Propranolol	Tin	65	1.79 (1.40–2.28)	6	2.31 (1.03–5.17)	no
Quinine	Tin	35	7.11 (5.09–9.94)	10	23.00 (11.99–44.15)	yes
Rofecoxib	Tin	131	2.53 (2.13–3.02)	8	2.26 (1.12–4.54)	no
Sildenafil	Tin	214	2.28 (1.99–2.61)	12	2.59 (1.46–4.59)	yes
Streptomycin	Tin	3	3.37 (1.08–10.51)	14	10.88 (6.31–18.74)	yes
Telmisartan	Tin	58	3.22 (2.49–4.17)	8	2.02 (1.00–4.07)	yes
Terbinafine	Tin	58	2.95 (2.28–3.82)	7	2.87 (1.36–6.07)	yes
Tobramycin	Tin	144	6.04 (5.12–7.12)	7	6.50 (3.06–13.80)	yes
Topiramate	Tin	193	2.68 (2.32–3.09)	9	1.99 (1.03–3.84)	yes
Tozinameran	Tin	56	6.53 (5.01–8.50)	171	10.11 (8.46–12.07)	no
Tramadol	Tin	183	1.46 (1.26–1.69)	10	1.99 (1.07–3.72)	yes
Trazodone Hydrochloride	Tin	59	1.33 (1.03–1.72)	12	4.63 (2.61–8.20)	yes
Valsartan	Tin	347	1.43 (1.29–1.59)	14	2.23 (1.31–3.80)	yes
Vancomycin	Tin	90	1.68 (1.37–2.07)	9	2.03 (1.05–3.92)	yes
Venlafaxine Hydrochloride	Tin	603	4.96 (4.57–5.38)	6	3.15 (1.40–7.05)	yes
Vortioxetine	Tin	84	2.74 (2.21–3.40)	4	3.06 (1.14–8.20)	yes

Table S6. Signals of previous screens available only from abstracts

Study	Scope	Reported	Crude	Adjusted	Replicated	Robust	Robust drugs
narasimhaswamy2026	tinnitus	4	0	0	0	0	
narasimhaswamy2026	tinnitus	3	2	1	1	1	IBUPROFEN
wang2024	tinnitus	3	2	2	2	2	CIPROFLOXACIN;DULOXETINE
wu2025	hearing_loss	5	3	2	0	0	